DLBAIPNLP01 - Project: NLP

Task 1: Sentiment Analysis on Movie Reviews

Arrehquette Ewube Eyong

IU International University

Table Of Content

Introduction

Sentiment analysis is an important application of Natural Language Processing (NLP) that involves detecting and classifying opinions found in text. Unlike traditional text classification tasks, sentiment analysis requires capturing the underlying meaning of text, which is very dependent on context (Paramesha et al., 2023). This project presents the development of a movie review sentiment analysis system capable of determining whether a review is either positive or negative.

This project develops a binary sentiment classifier using Linear Support Vector Machine (SVM) applied on Stanford IMDb movie review dataset. It follows a typical supervised NLP workflow including loading the dataset, preprocessing (lowercasing, HTML removal, stop word filtering), feature extraction through term frequency-inverse document frequency (TF-IDF). The model was progressively trained and evaluated on different size datasets. The performance of the Linear SVM is evaluated using metrics such as accuracy, precision, recall, F1-score and confusion matrix analysis.

The developed system demonstrates how NLP and machine learning techniques can be applied to automatically analyze public opinions from textual reviews. The system can also be extended to other domains such as product reviews, social media analysis and customer feedback systems.

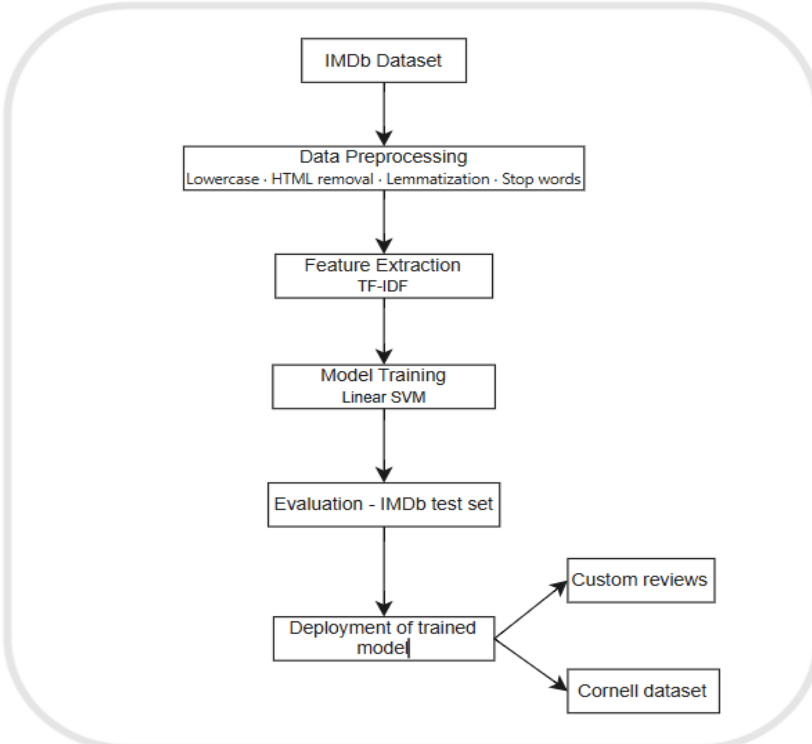

Figure 1: Project Workflow

Methodology

A supervised learning approach is used for binary sentiment classification using Linear Support Vector Machine (SVM). The methodology follows a standard NLP pipeline consisting of 6 main stages;

- Data collection
- Text preprocessing
- Feature extraction
- Model training
- Model evaluation
- Deployment

a. Data Collection/Preparation

The IMDb Large Movie Review Dataset is loaded directly from its directory structure, where the data is already split into training and testing sets. Each set contained a total of 25,000 movie reviews distributed across positive and negative categories. The positive were assigned the label 1 while the negative reviews were assigned 0. For both the training and testing dataset, the positive and negative reviews were combined into a single dataset and shuffled to remove any ordering bias.

b. Data Preprocessing

Each review undergoes a sequence of cleaning steps to reduce noise, normalize the text and make the data suitable for the model. The following preprocessing techniques were applied:

- Conversion of all text to lowercase to ensure consistency
- Removal of non-alphabetic characters, including number, HTML tags, punctuations
- Tokenization
- Removal of stop words like “the”, “is” and “and”
- Reducing words to their base dictionary form using lemmatization. This approach not only reduces noise resulting from variability but also captures more abstract affective meaning of a sentence (Babanejad et al., 2024)

These preprocessing steps were implemented using the Python libraries re, nltk. After this cleaning, the words are joined back into complete sentences and ready for feature extraction.

c. Feature Extraction

After preprocessing, the cleaned text data was transformed into numerical feature vectors using the Term Frequency-Inverse Document Frequency (TF-IDF) technique. TF-IDF measures the importance of a word in a document relative to the entire dataset. Terms that occur frequently within a text receive higher importance, whereas terms that appear frequently across many documents are assigned lower importance (Cahyanti et al., 2020). To handle mixed sentiment reviews, TF-IDF assigns higher weights to terms that carry stronger and more frequent sentiment signals, allowing the model to lean towards the dominant sentiment in the review.

The TfidfVectorizer was implemented with a maximum of 25,000 features and an n-gram range of 1 to 3. This allowed the model to consider single words (unigrams), pairs of consecutive words(bigrams) and sequences of three words(trigrams) during feature extraction.

d. Model Training

The Linear Support Vector Machine (SVM) classifier was used as the machine learning model for sentiment classification. The Linear SVM was trained on the TF-IDF feature matrix with a regularization parameter of C=1. The model works by finding a decision boundary that best separates the positive and negative reviews within the high-dimensional feature space.

The Linear SVM was chosen because TF-IDF generates high dimensional feature spaces that are linearly separable, making it effective for sentiment classification. So, the Linear SVM will perform better and computationally efficient. The low regularization parameter C =0.5 was selected to allow a wider soft margin which tolerate some misclassifications and improve the model's generalization.

e. Evaluation/Results

The Linear SVM achieved an overall accuracy of 87.5% on the test dataset.

The detailed classification report is presented in Table 1 below

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.87	0.89	0.88	12500
Positive (1)	0.88	0.86	0.87	12500

Weighted Average	0.88	0.88	0.88	25000

Table 1: Classification Report for the Linear SVM on the IMDb test set

From Table 1 above, the negative class achieved a precision of 0.87, meaning that 87% of the reviews predicted as negative were indeed negative. Its recall of 0.89 indicates the model correctly identified 89% of all actual negative reviews. The F1 score of 0.88 show an overall strong performance between the precision and recall.
For the positive class, the model achieved a precision of 0.88, meaning 88% of reviews predicted as positive were truly positive. The recall of the 0.86 show that the model captured 86% of all actual positive reviews. All this got an F1 score of 0.87.
The weighted average values of 0.88 for precision, recall and F1 score indicate that the model performed consistently across the entire test dataset of 25,000 reviews.
Overall, these results show that the Linear SVM model was highly effective for binary classification task of the textual movie review data.

The confusion matrix provides numeric details of the classification report, showing the exact number of correct and incorrect predictions for each class

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	11,076	1,424
Actual Positive (1)	1,690	10,810

Table 2: Confusion Matrix – IMDb test set

The confusion matrix in Table 2 shows that the model correctly classified 11,076 negative reviews (true negatives) and 10,810 positive reviews (true positives). This accounts to majority of predictions. The total number of right classification amount to 21,886, which is consistent with the report accuracy of 87.5%

On the other hand, 1,424 negatives reviews were incorrectly predicted as positive (false positive) and 1,690 positive reviews were incorrectly predicted as negative (false negatives).

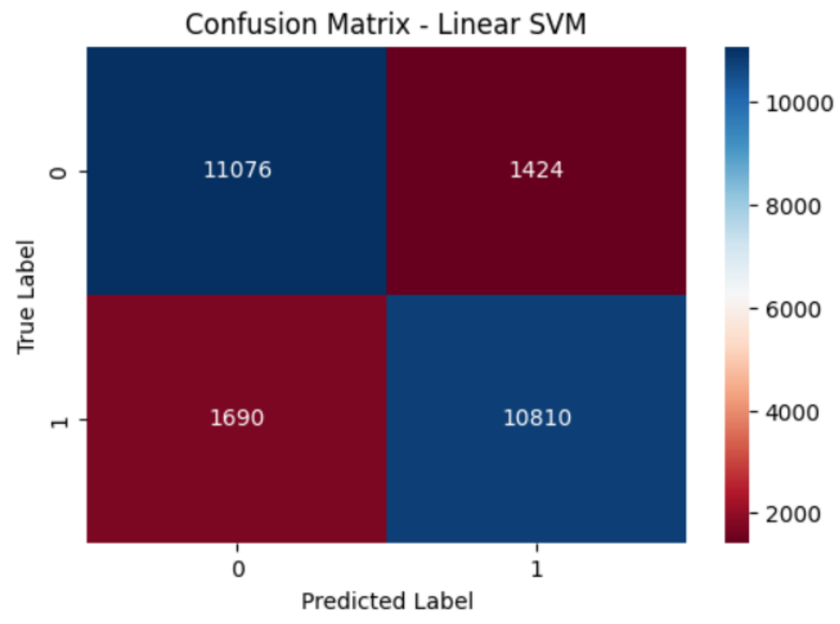

Figure 2: Confusion Matrix – IMDB Dataset

To further evaluate the model, a Receiver Operating Characteristics (ROC) is used. It was built on the confusion matrix: true positive rate vs. false positive rate. The Area Under the Curve (AUC) was computed. AUC measures the probability that a classification model will rank a randomly selected positive example higher than a randomly selected negative example (Fawcett, 2006).

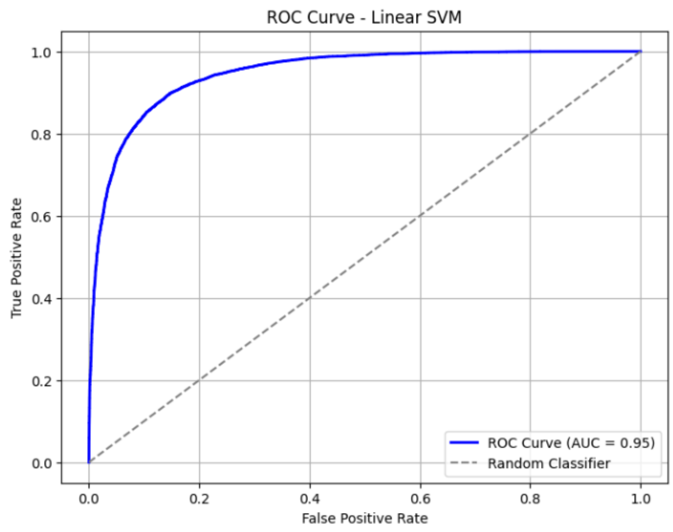

Figure 3: ROC Curve – Linear SVM (IMDB Dataset)

As seen in Fig.2 above, the ROC curve illustrates the trade-off between the True Positive Rate and the False Positive Rate across all possible decision thresholds. The AUC of 0.95 reflects that the model correctly distinguishes a positive review from a negative review 95% of the time. The high AUC confirms that the Linear SVM maintains a strong discriminative performance across all decision thresholds, further supporting the effectiveness of the overall pipeline.

The model used Hinge loss as its loss function, which penalizes misclassified samples according to their distance from the decision boundary. The model obtained a hinge loss of 0.38, suggesting that the majority of reviews were correctly classified and found beyond the margin.

f. Deployment

The trained Linear SVM model was then applied on set of custom movie reviews to see how practical the trained model is beyond the test dataset.

Each unseen written reviews were passed through the same preprocessing pipeline during training (lowercasing, HTML tag removal, non-alphanumeric removal, tokenization, lemmatization, stop word removal) to ensure the consistency of the data. The cleaned data was then fitted into a TF-IDF vectorizer for feature extraction and then fed into the trained Linear

SVM model for binary classification.
The following results were obtained

Review	Predicted Sentiment
"This movie was absolutely amazing. I loved every part of it!"	Positive Review
"The movie was boring and too long. I almost fell asleep."	Negative Review
"Terrible film. Waste of time and money"	Negative Review
"Great visuals, but the story was dull and unconvincing."	Negative Review
"Slow beginning, but a truly moving and memorable ending."	Positive Review

Table 3: Model prediction on custom sample reviews

As seen in table 3 above, the predictions align well with the expected sentiments. The clearly positive reviews like – "This movie was absolutely amazing. I loved every part of it" and the clear negative reviews like- "Terrible film waste of time and money", were both accurately predicted. For fixed sentiment reviews like – "Great visuals, but the story was dull and unconvincing", the negative conclusion outweighed the initial positive observation at the beginning of the sentence. And vice versa with the last review.
These results suggest the model is capable of generalizing on unseen dataset beyond the IMDb dataset, including reviews with both positive and negative opinions.
To further evaluate the model, the trained model was applied to the Cornell Movie Review Dataset, which contains 2000 movie reviews evenly divided between positive and negative sentiments. The model achieved an overall accuracy of 86.1% on this Cornell dataset. The detailed classification report is presented in Table 4 below

Class	Precision	Recall	F1-Score	Support
Negative (0)	0.85	0.87	0.86	1000
Positive (1)	0.87	0.85	0.86	1000
Weighted Average	0.86	0.86	0.86	2000

Table 4: Classification Report - Cornell Movie Review Dataset

From Table 4 above, the model achieved balanced performance across both classes with a F1-score of 0.86 for both classes indicating the model generalized well on unseen data and remains unbiased towards both sentiment category

The model correctly classified 872 negative and 850 positive reviews, giving a total of 1722 correct predictions out of 2000. It predicted 128 negatives as positive and 150 positives as negatives making a total of 278 misclassified reviews as seen in Table 5 below

	Predicted Negative (0)	Predicted Positive (1)
Actual Negative (0)	872	128
Actual Positive (1)	150	850

Table 5: Confusion Matrix – Cornell Movie Review Dataset

The relatively small number of misclassifications suggest the model is effective as distinguishing between positive and negative sentiments, although some reviews contain mixed or unclear expressions.

Conclusion

The project successfully developed a binary sentiment analysis system using supervised machine learning techniques for classifying movie reviews into positive and negative categories. The pipeline included all key steps such as text preprocessing, TF-IDF feature extraction using unigrams, bigrams and trigrams, followed by model training, evaluation and testing on unseen data. The complete pipeline included data collection, text preprocessing, feature extraction through TF-IDF, Linear SVM model training and model evaluation on IMDb and Cornell Movie Review datasets.

The model obtained an accuracy of 87.5% on the IMDB test set and 86.1% on the Cornell dataset, with balanced precision, recall and F1-scores. Additionally, the model handled the written mixed sentiment reviews by relying on the TF-IDF weighting, which assigned higher importance to the more dominant sentiment in the review

Overall, the results of this project show that the TF-IDF feature extraction and Linear SVM classifier provide a reliable approach for sentiment analysis of movie reviews.

List of Figures and Tables

Tables

Figures

Github Repository

Reference

- Paramesha, K., Gururaj, H. L., Nayyar, A., & Ravishankar, K. C. (2023). Sentiment analysis on cross-domain textual data using classical and deep learning approaches. Multimedia Tools and Applications, 82*, 30759–30782. https://doi.org/10.1007/s11042-023-14427-9it
- Tom Fawcett. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, *27*, 861–874. https://doi.org/10.1016/j.patrec.2005.10.010
- Cahyanti, F. E., Adiwijaya, & Faraby, S. A. (2020). On The Feature Extraction For Sentiment Analysis of Movie Reviews Based on SVM. *IEEE 2020 8th International Conference on Information and Communication Technology (ICoICT)* (pp. 1–5). https://doi.org/10.1109/ICoICT49345.2020.9166397
- Babanejad, N., Davoudi, H., Agrawal, A., An, A., & Papagelis, M. (2024). The Role of Preprocessing for Word Representation Learning in Affective Tasks. *IEEE Transactions on Affective Computing ; Volume 15, Issue 1, Page 254-272 ; ISSN 1949-3045 2371-9850*. https://doi.org/10.1109/taffc.2023.3270115